Current Ripple ≈ 0.8922 App

Voltage Ripple ≈ 7.965 mVpp

$$t_{on} = \frac{V_{out}}{V_{in} \cdot f_{sw}} = \frac{5V}{13.5V \cdot 750kHz} = 493.8\,nS$$

$$D = \frac{V_{out}}{V_{in}} = \frac{5V}{13.5V} = 0.37$$

$$\Delta I_L = \frac{(V_{in} - V_{out}) \cdot D}{L \cdot f_s} = \frac{(13.5V - 5V) \cdot 0.37}{4.7\mu H \cdot 750kHz} = 0.8922\,A_{pp}$$

ESR @ 22µF = 4.5mΩ

DC Bias % = 0.66 → cap datasheet graph (see below)

$C_{eff} = 22\mu F(1 - 0.66) = 7.48\mu F$, $C_{out} = 3 \cdot 7.48 = 22.44\,\mu F$

ESR_{tot}: $\frac{1}{ESR_{tot}} = \frac{1}{ESR} + \frac{1}{ESR} + \frac{1}{ESR} = \frac{3}{ESR} = \frac{3}{4.5}$ $ESR_{tot} = \frac{4.5}{3} = 1.5m\Omega$

$$V_{Ripple} = \Delta I_L \left(ESR + \frac{1}{8 \cdot f_{sw} \cdot C_{out}}\right) = 0.8922\left(1.5m\Omega + \frac{1}{8 \cdot 750kHz \cdot 22.44\mu F}\right) \approx 7.965\,mV_{pp}$$

Electrical Characteristics Data

KYOCERA

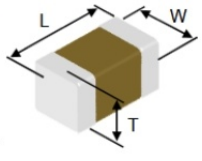
0603 / X5R / 22µF / +/-20% / 16V : KGM15CR5~~1C226M~~ (CM105X5R226M16A)

Dimension

1608(JIS) / 0603(EIA)

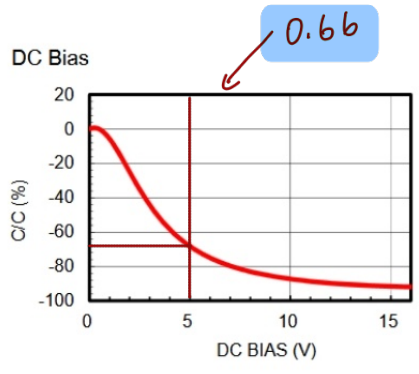
Unit:mm

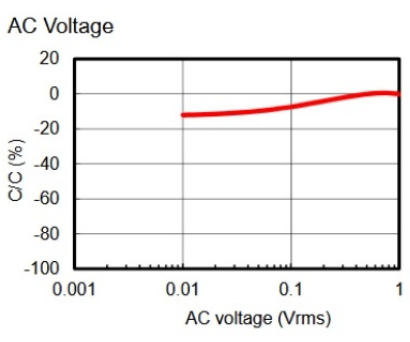
L	1.6±0.2
W	0.8±0.2
T	0.8±0.2

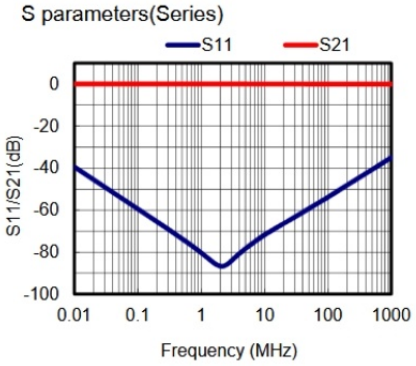


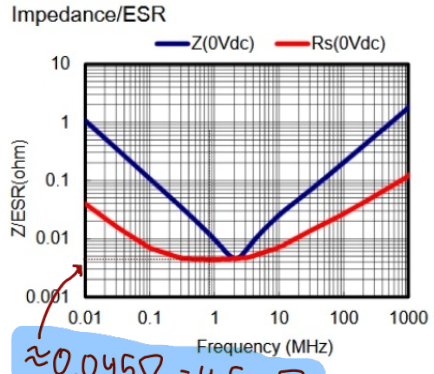
Characteristic

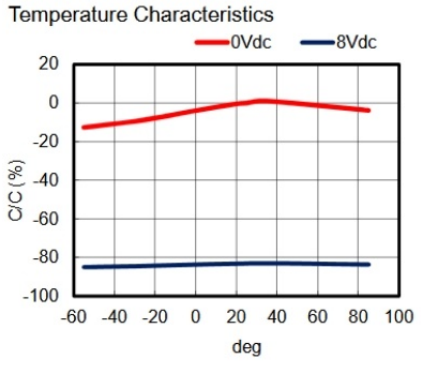
Temp. Range	-55deg to85deg
Temp. coeff	+/-15%
Capacitance Tolerance	22µF +/-20%
Rated Voltage	16Vdc











Capacitor Division